COMPLETE SYSTEM & FLOW BLUEPRINT

AI Personal Career Counselor

End-to-End System Requirements Mapped Directly to the 6-Stage Student Journey

CORE PROCESS MODEL	SYSTEM ARCHITECTURE	MATCHING STRATEGY	STATUS
One Continuous Guided Flow	React 19 + FastAPI + Supabase	Hybrid Multi-Factor (1,016 O*NET)	Verified Production Blueprint

1. Executive Summary & Flow Philosophy

The **AI Personal Career Counselor** transforms college career guidance from a disconnected one-off questionnaire into a rigorous, continuous journey. It operates across two distinct pillars:

- **Pillar 1: Counselor Mode (Discovery & Validation)** — Uncovers student identity via 24-item Holland RIASEC psychometrics, subconscious values ("Know Me"), resume skill extraction, and dense semantic vector matching across 1,016 O*NET occupations.
- **Pillar 2: Coach Mode (Execution & Habit Formation)** — Once the student commits to a goal, unlocks personalized roadmaps, NLP skill-gap pinpointing, dynamic daily scheduling, and an AI career coach.

> **The Fundamental Product Rule: Choose First, Then Plan**
> The platform strictly prevents premature dashboard access. Every student must complete Onboarding and Assessment, and explicitly commit to an active career goal before Phase 2 coach modules unlock.

2. High-Level Lifecycle Progression Summary

STAGE	STEP NAME & ROUTE	PRIMARY COMPONENT	DATABASE MILESTONE STATE	NEXT PERMITTED ACTION
0	**Auth & Hydration** (`/login`)	`ProtectedRoute.tsx` / `AuthContext`	Active Supabase session + profile resolved	Resume incomplete stage
1	**Onboarding** (`/onboarding`)	`Onboarding.tsx`	`profiles` record written	Navigate to `/assessment`
2	**Assessment** (`/assessment`)	`Assessment.tsx` (RIASEC + Know Me)	`assessments` record written	Score → View recommendations
3	**Hybrid Match** (`/api/career-match`)	FastAPI `matcher.py` + Groq AI	`assessments.matches` cached	Explore & compare careers
4	**Goal Commitment** (`career_goals`)	`Assessment.tsx` Confirmation Modal	`career_goals` row (`status='active'`)	Unlock `/dashboard` & Phase 2
5	**Career Mode** (`/dashboard` , `/today`)	`Dashboard.tsx` / `Roadmap.tsx`	`roadmaps` & `task_progress`	Daily tasks, skill-gap & roadmap

3. Step-by-Step Requirements Mapped Directly to the Project Flow

0 Stage 0: Authentication, Session Initialization & Route Hydration

Completed & Secured

User Action: Student registers or signs in using Email/Password or Google OAuth.

PRIMARY ROUTES	DATABASE STATE	SECURITY & PROTECTION
`/login`, `/signup`, `ProtectedRoute`	Supabase `auth.users` session	Supabase JWT Bearer Tokens

System Requirements & Implementation Rules:

- **Defensive Hydration:** `ProtectedRoute.tsx` must explicitly wait until `authLoading == false` and `profileLoading == false` before triggering any routing decisions, completely eliminating infinite redirect loops.
- **Resume-on-Login:** Returning users must resume at their exact incomplete milestone (e.g. unassessed students resume at `/assessment`, unconfirmed students resume at `/assessment?step=results`) rather than being blindly dumped onto the home or dashboard page.
- **Tenant Isolation:** Every downstream request authenticates using Supabase JWT tokens enforced by Row Level Security (RLS).

1 Stage 1: Academic Onboarding & Optional Resume Parsing

Verified Flow

User Action: Student inputs educational background and optionally drags-and-drops their PDF resume.

PRIMARY ROUTE	DATABASE STATE	BACKEND API
`/onboarding`	Supabase `profiles` record written	`POST /api/upload-resume`

System Requirements & Implementation Rules:

- **Core Academic Inputs:** Must collect educational tier (Degree, Diploma, PUC/12th, SSLC/10th), institution name, course/branch, academic year, personal interests, and weekly study hours budget.
- **Target Role Optionality:** `targetRole` is captured as non-binding background context only. Students are never forced to finalize a career choice during onboarding.
- **Optional Resume Parsing:** Accepts PDF files (max 5 MB). PyMuPDF extracts text; Groq LLM (`openai/gpt-oss-120b`, JSON mode) parses technical skills, education, and experience. Operates with deterministic regex fallback if offline.
- **Strict Forward Navigation:** Upon clicking "Continue to Career Discovery", the application saves data to `profiles` and navigates strictly to `/assessment`. Premature redirects to `/dashboard` are eliminated.

2 Stage 2: Psychometric & Subconscious Identity Assessment

Verified Flow

User Action: Student completes the 24-item RIASEC psychometric test followed by the "Know Me" identity discovery module.

PRIMARY ROUTE	DATABASE STATE	BACKEND APIS
`/assessment`	Supabase `assessments` row created	`GET /api/assessment/questions`, `POST /api/assessment/score`

System Requirements & Implementation Rules:

- **24-Question Holland Code (RIASEC):** Deterministic mathematical scoring calculates 0–100% scores across Realistic, Investigative, Artistic, Social, Enterprising, and Conventional traits, extracting the primary 3-letter Holland code (e.g. `IRC`).
- **"Know Me" Subconscious Profiling:** Uncovers authentic career drivers via:
  1. *Equal-Salary Test* — identifies intrinsic interest when pay is identical.
  2. *Energy Source Audit* — evaluates tasks that energize vs. drain the student.
  3. *Secret Envy Compass* — diagnoses genuine unconscious aspirations.
  4. *Anti-Goal Dealbreakers* — non-negotiable boundaries (e.g., no night shifts, no coding, relocation budget constraints).
- **State Resilience:** Answers persist in component state and rehydrate cleanly across browser refreshes.

3 Stage 3: Multi-Factor Hybrid Matching & Career Exploration

Verified Pipeline

User Action: Student explores the Top 5 recommended careers, reviews Indian salary tiers, daily life, and compares options.

PRIMARY ROUTES	DATABASE STATE	BACKEND API
`/assessment?step=results`, `/careers/:slug`	`assessments.matches` JSON cached	`POST /api/career-match`

System Requirements & Implementation Rules:

- **Hybrid Multi-Factor Scoring Formula:**

```
score = (0.50 * vector_similarity) + (0.30 * riasec_fit) + (0.20 * skill_overlap) - dealbreaker_penalties
```

  Evaluated across 1,016 real O*NET occupational dense embeddings (`all-MiniLM-L6-v2`).
- **Dealbreaker Deduction:** Deducts `0.35` per conflict with student anti-goals, ensuring unviable roles are filtered out.
- **Dual-Track AI Synthesis:**
  - *Track A (Primary):* Groq LLM (GPT-OSS 120B) reranks the top 15 candidates into a refined Top 5, providing personalized fit reasons and watch-outs.
  - *Track B (Fallback):* If Groq times out or is rate-limited, deterministic rule-based ranking completes immediately without user disruption.
- **In-Depth Exploration & Comparison:** Career detail pages (`/careers/:slug`) show Indian salary curves (Fresher/Mid/Senior), typical day-to-day routines, and 3-way side-by-side comparison modal.

4 Stage 4: Deliberate Career Goal Commitment & Atomic Persistence

Task 2 Verified

User Action: Student reviews recommendations and confirms their choice via the "Make this my goal" commitment dialog.

PRIMARY MODAL	DATABASE STATE	DATABASE CONSTRAINT
Career Confirmation Dialog	`career_goals (status='active')`	`career_goals_one_active` partial unique index

System Requirements & Implementation Rules:

- **Reusing Contract:** Leverages the verified `confirmCareerGoal(title, slug, field)` implementation.
- **Database-Level Integrity:** PostgreSQL unique partial index guarantees exactly ONE active goal per student:

```
CREATE UNIQUE INDEX career_goals_one_active ON career_goals(user_id) WHERE status = 'active';
```

  When changing goals, previous goals transition to `status = 'archived'` while retaining historical task progress.
- **Atomic Profile Synchronization:** Concurrently writes to `career_goals` and updates `profiles.target_role`.
- **Visible Failure Handling:** If network or Supabase persistence fails, displays a recoverable error toast and prevents navigation to `/dashboard`.

5 Stage 5: Career Mode Transition, Route Authorization & Daily Execution

Task 3 Route Guarded

User Action: Student enters the personalized dashboard and executes their daily tasks toward job readiness.

PRIMARY ROUTES	DATABASE STATE	BACKEND APIS
`/dashboard`, `/today`, `/roadmap`, `/skills`	`roadmaps`, `task_progress`	`POST /api/skill-gap`, `POST /api/roadmap/generate`

System Requirements & Implementation Rules:

- **Global Career Mode Personalization:** Greets user dynamically ("Good morning, future Data Scientist 🌱"), with field-specific themes and illustrations.
- **NLP Skill-Gap Analysis:** Compares resume competencies against occupational benchmarks, outputting *Have*, *Partial*, and *Missing* competencies along with a quantitative % job-readiness score.
- **Phased Learning Roadmap:** Automatically generates hierarchical milestones (4 Worlds × 3–6 Levels) adapted to graduation year and available study hours.
- **Dynamic Daily Task Rollover:** The daily scheduler packs 2–3 tasks for today. Missed tasks automatically roll forward to prevent demotivating overdue backlogs.

4. Complete Journey Route Guard Rules Matrix

The centralized routing engine in `ProtectedRoute.tsx` enforces strict stage prerequisites to ensure students cannot bypass required steps via direct URL manipulation:

STUDENT STAGE	DATABASE MILESTONE STATE	ATTEMPTED URL ROUTE	ENFORCED ROUTING ACTION
Unauthenticated	No active Supabase auth session	Any private route	Redirect to `/login` (saves target in router state)
Incomplete Onboarding	`hasCompletedOnboarding(profile) == false`	`/assessment`, `/dashboard`, `/roadmap`	Redirect to `/onboarding`
Incomplete Assessment	`onboarding complete`, but `latestAssessment == null`	`/dashboard`, `/roadmap`, `/today`	Redirect to `/assessment`
Goal Not Confirmed	`assessment complete`, but no active `career_goals`	`/dashboard`, `/roadmap`, `/today`	Redirect to `/assessment?step=results`
Fully Committed	Active goal in `career_goals` and matching `target_role`	`/dashboard`, `/today`, `/roadmap`, `/skills`	Full Access Authorized

5. Relational Database Schema & RLS Policies (Supabase PostgreSQL)

All student data tables enforce strict multi-tenant isolation using PostgreSQL Row Level Security (RLS) policies:

```
-- 1. Profiles Table
CREATE TABLE profiles (
  user_id UUID PRIMARY KEY REFERENCES auth.users ON DELETE CASCADE,
  full_name TEXT, college TEXT, course TEXT, year TEXT,
  interests TEXT[] DEFAULT '{}', hours_per_week INT DEFAULT 10,
  target_role TEXT, resume JSONB,
  created_at TIMESTAMPTZ DEFAULT NOW(), updated_at TIMESTAMPTZ DEFAULT NOW()
);
ALTER TABLE profiles ENABLE ROW LEVEL SECURITY;
CREATE POLICY "Users access own profile" ON profiles
  FOR ALL USING (auth.uid() = user_id) WITH CHECK (auth.uid() = user_id);

-- 2. Assessments Table
CREATE TABLE assessments (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID NOT NULL REFERENCES auth.users ON DELETE CASCADE,
  answers JSONB NOT NULL, scores JSONB NOT NULL, code TEXT NOT NULL,
  matches JSONB, source TEXT CHECK (source IN ('ai', 'rule')),
  created_at TIMESTAMPTZ DEFAULT NOW()
);
ALTER TABLE assessments ENABLE ROW LEVEL SECURITY;
CREATE POLICY "Users access own assessments" ON assessments
  FOR ALL USING (auth.uid() = user_id) WITH CHECK (auth.uid() = user_id);

-- 3. Confirmed Career Goals Table
CREATE TABLE career_goals (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID NOT NULL REFERENCES auth.users ON DELETE CASCADE,
  career_slug TEXT NOT NULL, career_title TEXT NOT NULL, field TEXT NOT NULL,
  status TEXT NOT NULL DEFAULT 'active' CHECK (status IN ('active', 'archived')),
  confirmed_at TIMESTAMPTZ DEFAULT NOW()
);
ALTER TABLE career_goals ENABLE ROW LEVEL SECURITY;
CREATE POLICY "Users access own goals" ON career_goals
  FOR ALL USING (auth.uid() = user_id) WITH CHECK (auth.uid() = user_id);

-- Enforce exactly one active goal per student
CREATE UNIQUE INDEX career_goals_one_active ON career_goals(user_id) WHERE status = 'active';
```

6. REST API Specifications & Contracts

METHOD	ENDPOINT	AUTH	REQUEST / RESPONSE CONTRACT & BEHAVIOR
`GET`	`/`	Public	Health check. Returns `{ status: "ok", ai_enabled: true, model: "gpt-oss-120b" }`.
`GET`	`/api/careers`	Public	Returns list of career taxonomy titles, slugs, and industry fields. Supports `?search=` and `?field=`.
`GET`	`/api/careers/{slug}`	Public	Detailed career profile: Indian salary tiers (Fresher/Mid/Senior), daily routine, key skills, pros/cons, and similar careers.
`POST`	`/api/upload-resume`	Bearer	Accepts `multipart/form-data` (max 5 MB PDF). Returns structured JSON: `{ name, education[], skills[], experience[] }`.
`GET`	`/api/assessment/questions`	Public	Returns 24 psychometric question objects: `{ id, text, dimension: "R"|"I"|"A"|"S"|"E"|"C" }`.
`POST`	`/api/assessment/score`	Bearer	Input: `{ answers: { [id]: score } }`. Returns normalized 0–100 scores per dimension and 3-letter Holland code.
`POST`	`/api/career-match`	Bearer	Input: RIASEC scores, student background, skills, and "Know Me" values. Computes hybrid dense vector similarity + RIASEC Euclidean distance + skill overlap. Returns top 5 careers with custom match reasons and watch-outs.
`POST`	`/api/skill-gap`	Bearer	Input: `{ resume_skills: string[], target_career_slug: string }`. Maps skills against occupational requirements into `have`, `partial`, and `missing` arrays with a job-readiness percentage.
`POST`	`/api/roadmap/generate`	Bearer	Generates or retrieves cached study roadmap for the confirmed career goal, customized by student graduation year and hours.
`POST`	`/api/counselor/chat`	Bearer	Conversational counselor agent diagnosing student situation, academic tier, energy drivers, and dealbreakers.

API Authentication & Authorization Rules

All `/api/*` endpoints requiring student context validate the Supabase JWT token received in the `Authorization: Bearer <token>` header. Unauthenticated calls automatically receive an HTTP 401 Unauthorized response.

7. Verification Matrix & Test Evidence

All stages of the unified journey were audited and validated across dedicated automated test suites:

TEST SUITE	FOCUS AREA	RESULT	VERIFIED PRODUCTION BEHAVIOR
`onboarding-journey.test.ts`	Stage 1 Navigation	Passed (100%)	Completing onboarding immediately routes to `/assessment`; premature dashboard routing removed.
`goal-confirmation.test.ts`	Stage 4 Persistence	Passed (100%)	Confirmed career title and slug persist atomically to `career_goals` and sync with `profiles.target_role`.
`route-guards.test.ts`	Journey Guarding	Passed (100%)	Direct URL tampering strictly intercepted. Incomplete milestones redirect to respective steps without loop issues.
`journey-audit.test.ts`	End-to-End Journey	Passed (100%)	Complete 6-stage flow verified from initial account creation through daily task execution.
`ai-fallback.test.ts`	Fault Tolerance	Passed (100%)	Groq service outage simulation confirmed instant fallback to deterministic scoring without UI errors.

8. Implementation Status & Product Key Results

100%	1,016	< 150ms	0%
PHASE 1 COMPLETE	O*NET OCCUPATIONS	VECTOR SEARCH LATENCY	ROUTE LEAK RATE

Phase 2 Target Key Results

- **Assessment Completion:** ≥ 80% of registered students complete onboarding and RIASEC profiling.
- **Goal Conviction:** ≥ 95% of students explicitly confirm a career goal before accessing daily roadmap features.
- **Daily Habit Formation:** ≥ 70% weekly active retention via dynamic rollover tasks and streak motivation.

Architecture Review & Sign-Off: All Phase 1 modules (C0–C5) and student journey route guards (Tasks 1–4) are verified and in production readiness. Phase 2 (Coach modules M2–M9) will branch from the confirmed `career_goals` schema.